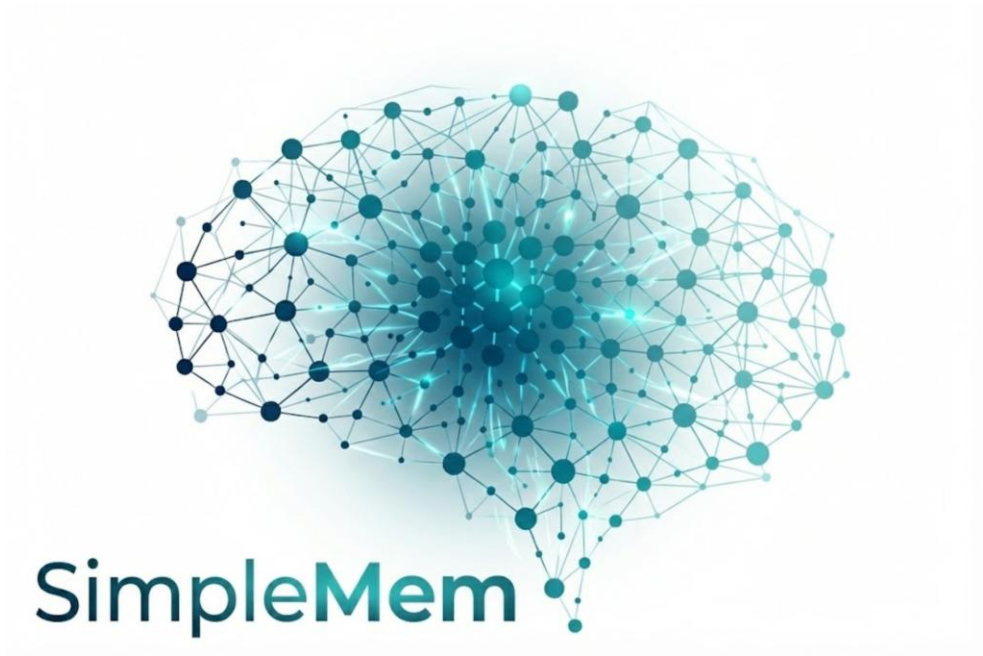
FTEC5660 Reproducibility Work

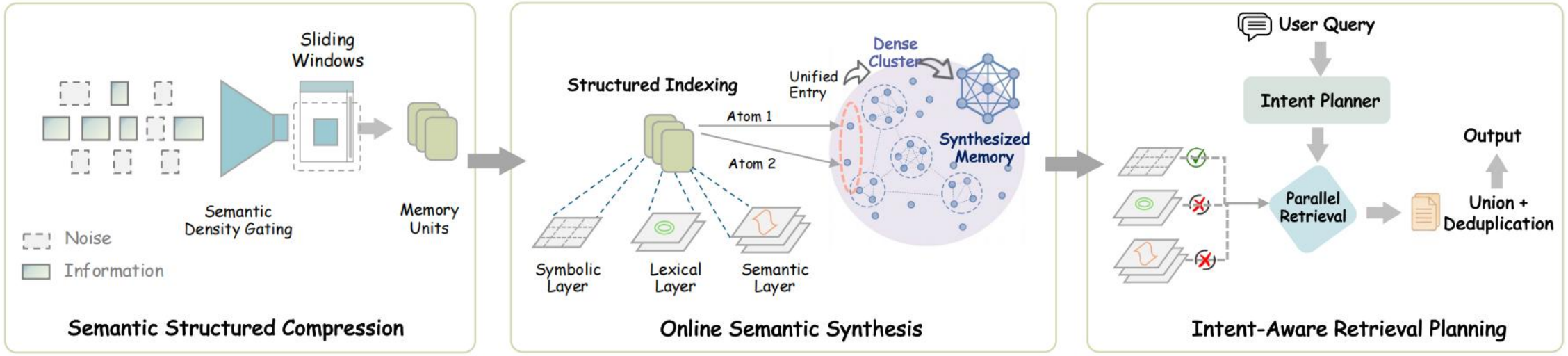
Reproducing SimpleMem

Does a second search improve answers enough to justify its cost?



The workflow I reproduced

SimpleMem builds searchable memories from long dialogues and retrieves them to answer questions.



Reproduction target **Run the upstream LoCoMo pipeline through scored answers.**

The reflection switch

Both runs independently plan and perform the initial hybrid search.

Reflection ON

Check whether the evidence is complete.
Search again if needed, then answer.

Reflection OFF

Generate the answer from the initial search results.

```
enable_reflection=reflection
```

True for the baseline, False for the ablation.

Paired experiment design

SimpleMem v0.1.0 with DeepSeek Flash and Qwen3 embeddings on CPU.

120

Fixed LoCoMo questions

3

Paired rounds

720

Scored answers

Matched inputs

Same questions, frozen memory snapshots, model, prompts and scorer in both settings.

Metrics

Macro token F1, retrieval-to-answer time and total API tokens per question.

Published and local baseline scores

The local run completed end to end and scored 0.91 F1 points below the paper.

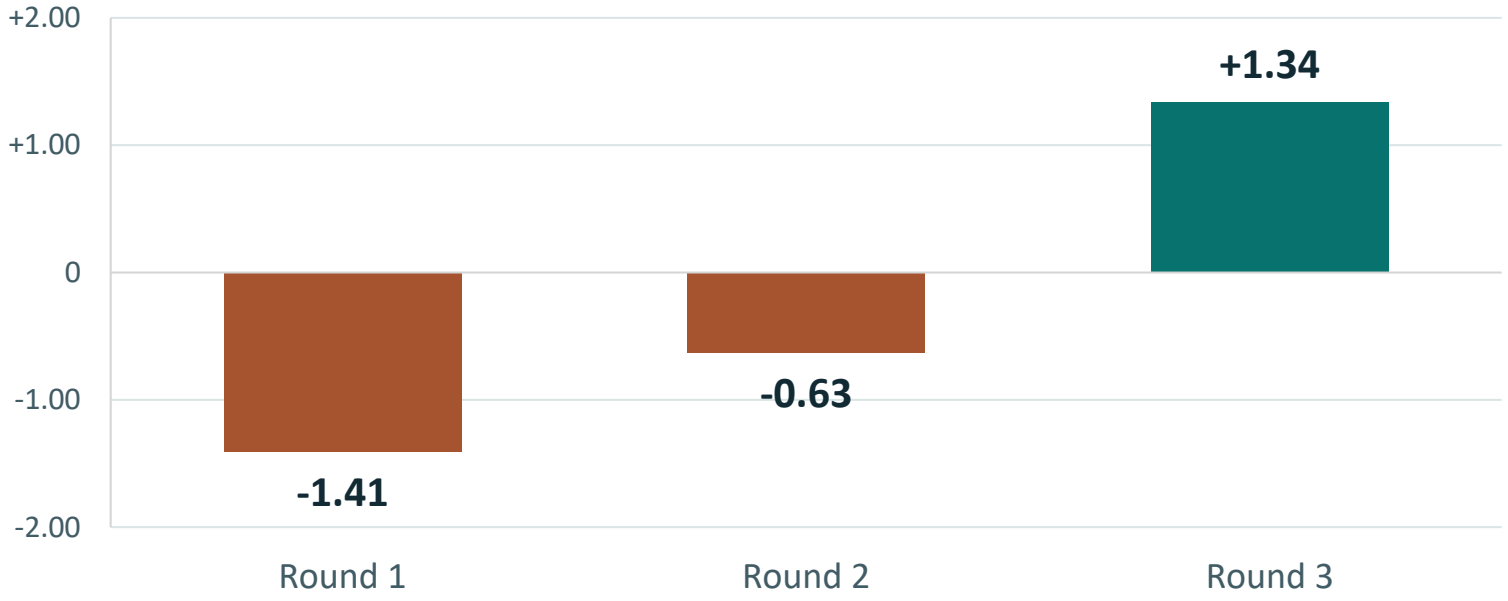
Measure	Published reference	Local baseline
Macro token F1	43.24	**42.33**
Language model	GPT-4.1-mini	**DeepSeek Flash**
Evaluation scope	LoCoMo-10	**120 questions across 3 conversations**
Memory window	20 turns	**40 turns**

The index configuration and category-level scores also differed.

The effect on F1 changed across rounds

Macro token F1 averaged 42.33 with reflection and 42.57 without it.

F1 difference in points (ON – OFF)



–0.23

Mean difference (points)

Across-round SD 1.41 points

Reflection increased latency and token use

Turning it off reduced both measures in every round.

Mean per question	Reflection ON	Reflection OFF	Reduction
Retrieval + answer time	14.26 s	8.15 s	**42.8%**
Total API tokens	13,392	4,454	**66.7%**
HTTP attempts	6.47	4.00	**38.1%**

Where the extra tokens went

Evidence checks and new query generation accounted for about 89%.

What the extra search retrieved

Among 179 pairs with added memories, F1 rose in 33, fell in 36 and tied in 110.

Round 3 examples	Tim’s most-visited country	John’s team trip
Reference answer	UK	a new city
New evidence	An earlier conference in the UK	Edinburgh came up as a suggestion
Token F1 (OFF to ON)	0.00 to 0.67	0.46 to 0.00

The UK pair started with different evidence.

The Edinburgh pair started with the same 31 memories in the same order.

What failed during reproduction

Dependency, index and query failures appeared at different stages.

Stage	Observed failure	Fix or outcome
Installation	LanceDB import failed	Pinned both namespace dependencies
Memory setup	Tantivy index write failed on Windows	Rebuilt in a fresh directory and verified keyword hits
Formal run	Tantivy rejected one question’s keyword query	Other channels retrieved evidence and all six responses were scored

The paired pilot confirmed both switch settings before the formal run.

Recommendations for a rerun

Verify each retrieval channel before spending API quota on the full evaluation.

Pin the inputs	Use the documented dependency, dataset and embedding versions.
Check a small pilot	Probe the keyword index and confirm that OFF makes zero reflection checks.
Preserve paired evidence	Reuse snapshots, log per-stage time and tokens, and inspect answer traces.

GitHub reproduction guide

Full report (PDF)

What reproduced and what remains open

Workflow	**SimpleMem built memories, retrieved evidence and scored 720 answers.**
Published score	**43.24 remains a reference under the paper’s original configuration.**
Reflection	**F1 differences changed sign. With reflection off, latency fell by 42.8% and API token use fell by 66.7%.**

For a similar workload, start with reflection off and test when a second search helps.